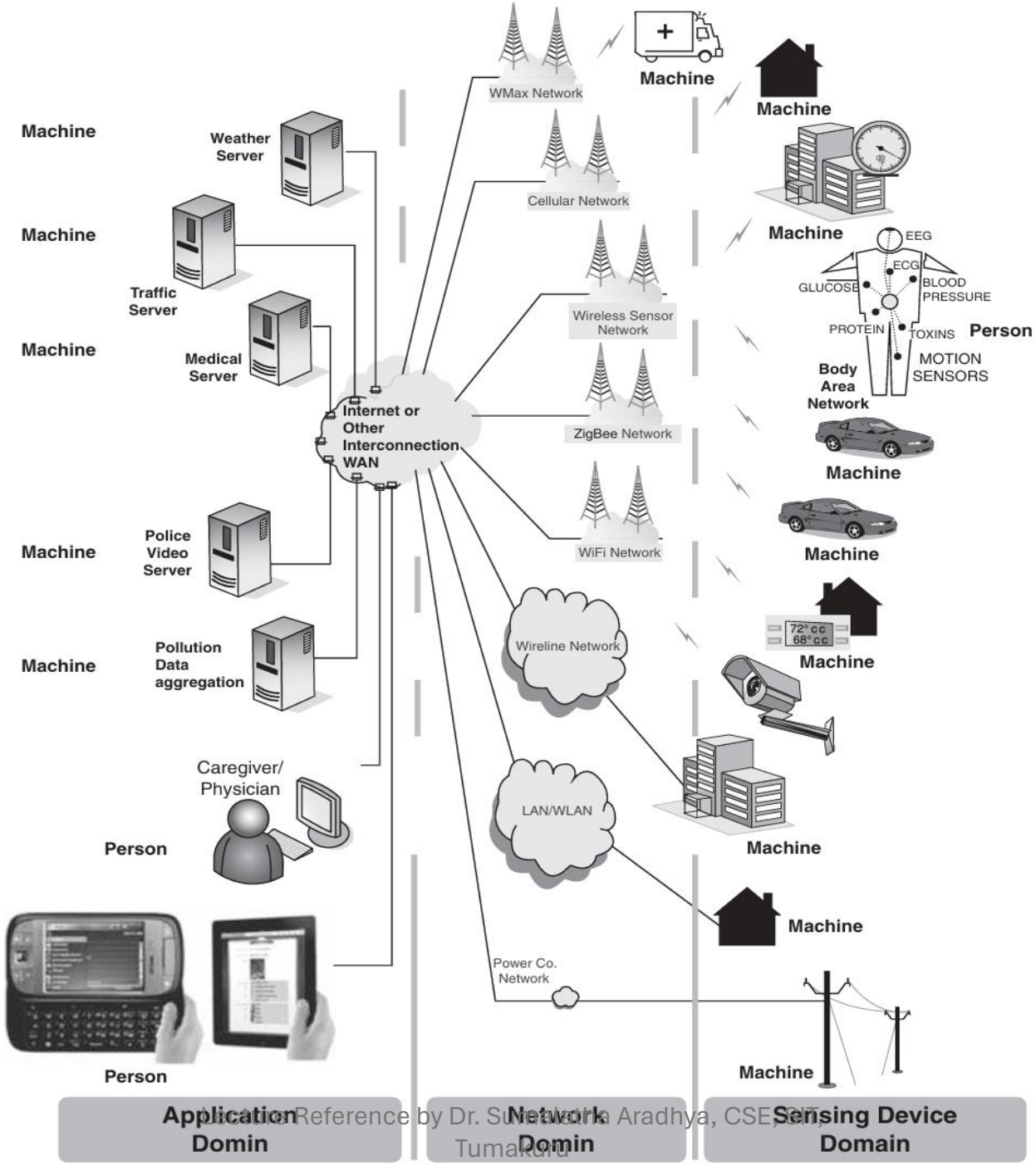

Lecture Reference by Dr. Sudarshana Aradhya, CSE, SIT, Tumakuru

- Standards drive economy-of-scale benefits for components;
- Standards also simplify the control and monitoring of patients in hospitals, in care facilities, and in homes
- e-Health and m-health rely on groups of connected devices, including devices communicating with classic smartphones using near field communication (NFC) technology, or with other nodes using low power, short-range radio communication technology such as Bluetooth low energy (BLE), or ZigBee , Kingsley

BLE is a low power version of Bluetooth capable of reporting data from a sensor for up to a year from a small button battery;

although the BLE data rate and radio range is lower than that of classic Bluetooth, also an IEEE standard, the low power and long battery life make it suitable for short-range monitoring applications in medicine

IEEE 802.15.1 (Bluetooth)

- ZigBee is designed for low-to-very-low-duty cycle static and dynamic environments with many active nodes;

- Bluetooth, on the other hand, is designed for high QoS, variety of duty cycles, and moderate data rates in networks with limited active nodes

Bluetooth is a WPAN technology based on IEEE 802.15.1.

It is a specification for short-range wireless connectivity for portable personal devices, including computer peripherals. It is now one of the most popular technologies in consumer electronics. Bluetooth was initially developed by Ericsson;

in the late 1990s, the Bluetooth Special Interest Group (SIG) made their specifications publicly available. Soon thereafter, the IEEE 802.15 Group took the Bluetooth work and developed a vendor-independent standard.

The Bluetooth SIG, in conjunction with the IEEE, has managed enhancements of the basic standard over the years.

Bluetooth has evolved through four versions

All versions of the Bluetooth standards maintain downward compatibility.

The Bluetooth SIG has approximately 17,000 member companies in telecommunication, computing, and CE.

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TABLE · · Versions of Bluetooth

Version	Description
Bluetooth v1.0 and v1.0B	Original versions; had limited interoperability
Bluetooth v1.1	This is original IEEE Standard 802.15.1–2002
Bluetooth v1.2	Ratified as IEEE Standard 802.15.1–2005. Incorporates a number of enhancements compared with v1.1 including (i) faster connection and discovery; (ii) use of AFH spread spectrum; (iii) supports higher transmission speeds up to 721 Kbps; and (iv) adds flow control mechanisms
Bluetooth v2.0 + EDR	Published in 2004. Incorporates a number of enhancements compared with v1.1 including faster data transfer of about 3 Mbps and lower power consumption through a reduced duty cycle. Note: To be exact, Version 2.0 devices have a higher power consumption; however, the fact that the transmission rate is three times faster (thereby reducing the transmission burst times), effectively reduces consumption to half that of 1.x devices
Bluetooth v2.1 + EDR	Published in July 2007. This release adds secure simple pairing (SSP), which improves the pairing process for Bluetooth devices while improving security; it also incorporates a subrating mechanism that reduces the power consumption in low-power mode
Bluetooth v2.1 + EDR	Published in July 2007. This release adds secure simple pairing (SSP), which improves the pairing process for Bluetooth devices while improving security; it also incorporates a subrating mechanism that reduces the power consumption in low-power mode
Bluetooth v3.0 + HS	Published in April 2009. This release supports a theoretical data transfer speeds of up to 24 Mbps by using the Bluetooth link for negotiation and establishment of a session for high data rate traffic carried over a collocated 802.11 link. It adds alternate MAC/PHY (AMP) for the use of 802.11 as a HS transport. Note: The HS portion of the specification is not mandatory, and only devices with the "+HS" label actually support the Bluetooth over 802.11 HS data transfer. The enhanced power control feature updates the power control feature to remove the open loop power control and also to clarify ambiguities in power control as related to EDR
Bluetooth v4.0	Published in June 2010. This version includes *Classic Bluetooth*, *Bluetooth high speed*, and BLE protocols. Bluetooth high speed is based on Wi-Fi and Classic Bluetooth consists of legacy Bluetooth protocols

: Bluetooth v4.1, 4.2, 5.0,5.1, 5.2, 5.3, 5.4

https://en.wikipedia.org/wiki/Bluetooth

medical implant communications service

Wireless Medical Telemetry Service

Ultra Wide Band

	MICS	WMTS	UWB (802.15.6)	IEEE 802.15.4 (ZigBee)	IEEE 802.15.1 (Bluetooth)	WLANs (802.11b/g)
Frequency band	402-405 MHz	608-614, 1395-1400, 1429-1432MHz	3-10 GHz	1.4 GHz (868/915MHz Eur./US)	2.4 GHz	2.4 GHz > 11 Mbps
Bandwidth	3 MHz	6MHz	>500 MHz	5 MHz	1 MHz	20 MHz
Data rate	19 or 76 kbps	76 kbps	850 kbps to 20Mbps	250 kbps (2.4 GHz)	721 kbps	
Multiple access	CSMA / CA, Polling	CSMA / CA, Polling	Not defined	CSMA / CA	FHSS/GFSK	OFDMA,CSMA/CA
Trans power	-16dbm (25μ W)	≥10dbm and <1.8 db	-41 dBm	0 dBm	4,20 dBm	250mW
Range	0-10	> 100 m	1,2 m	0-10 m	10,100 m	0-100 m

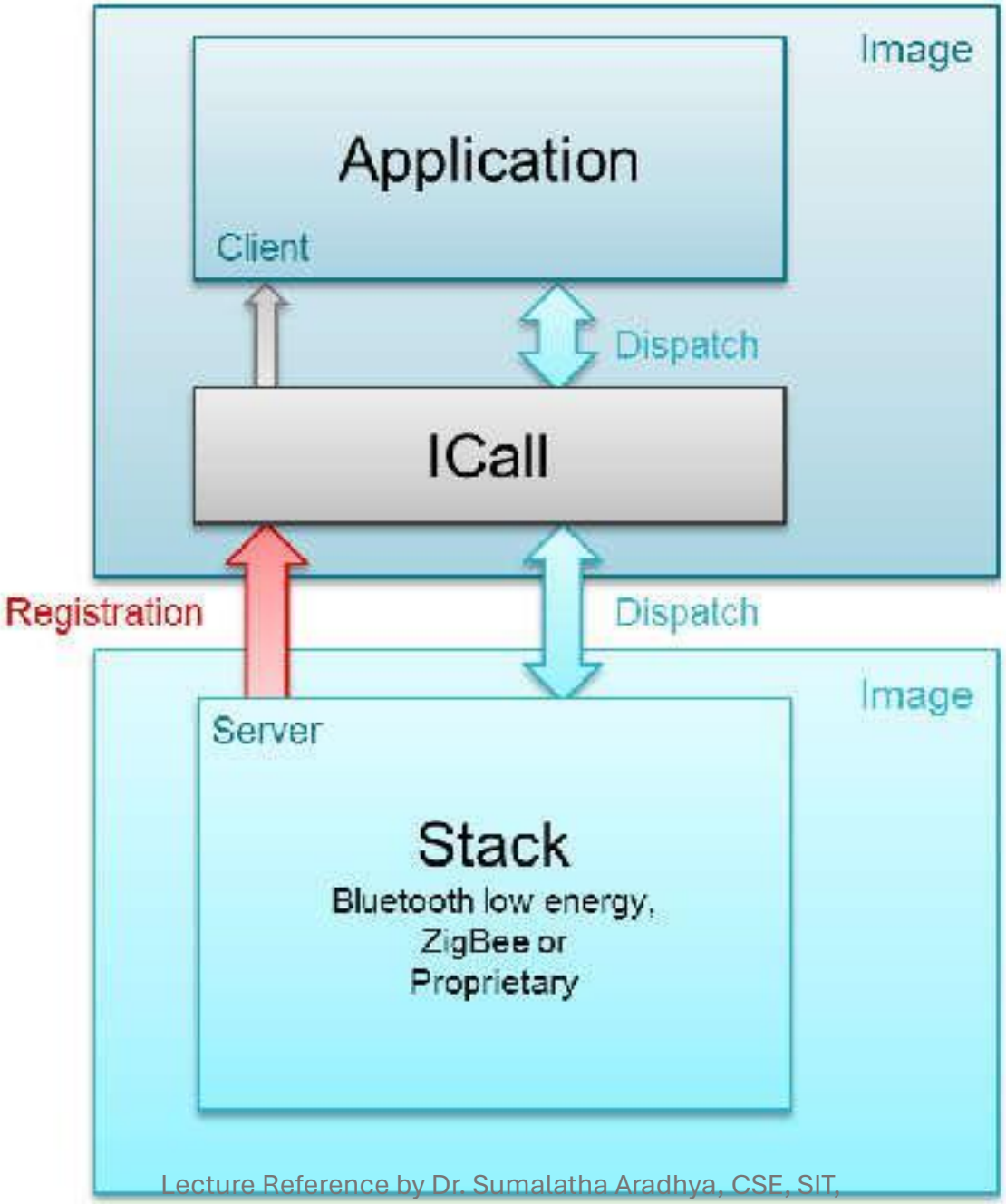
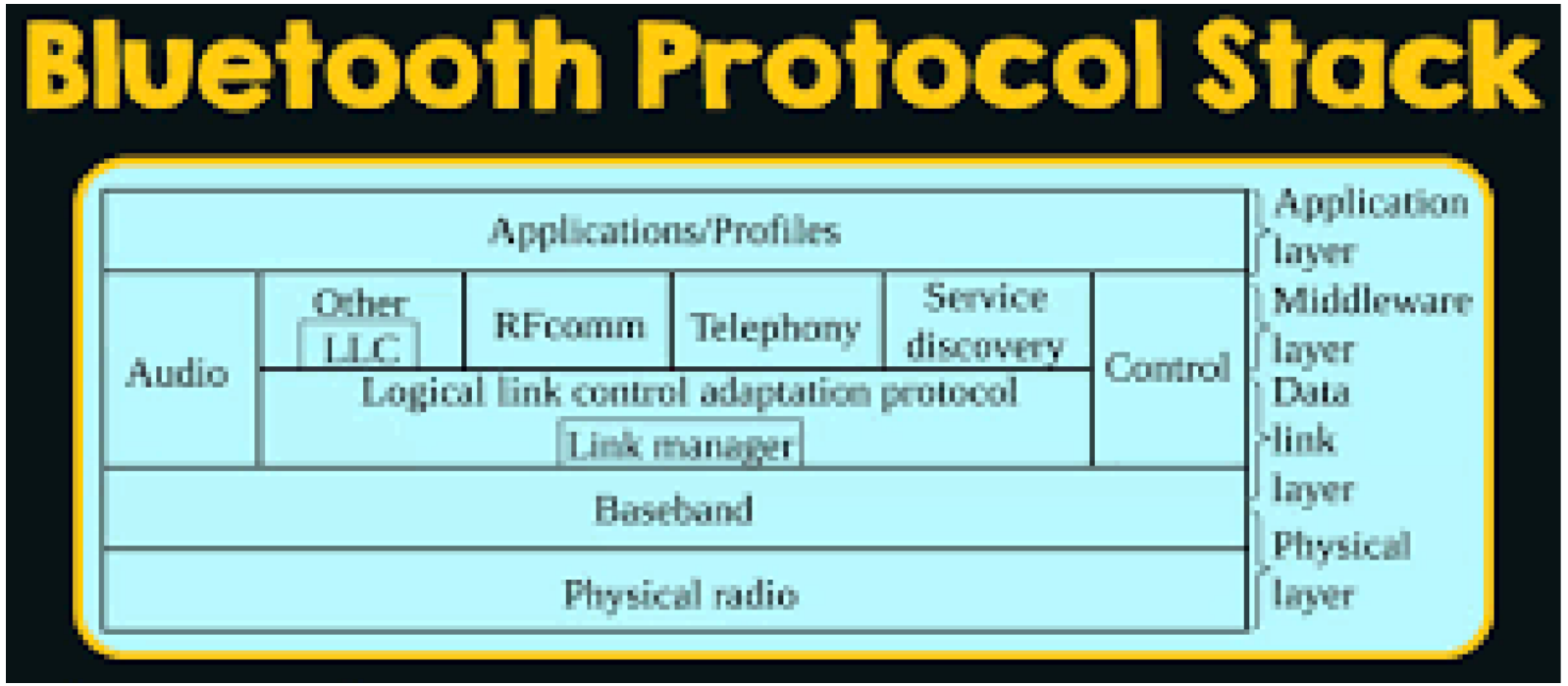
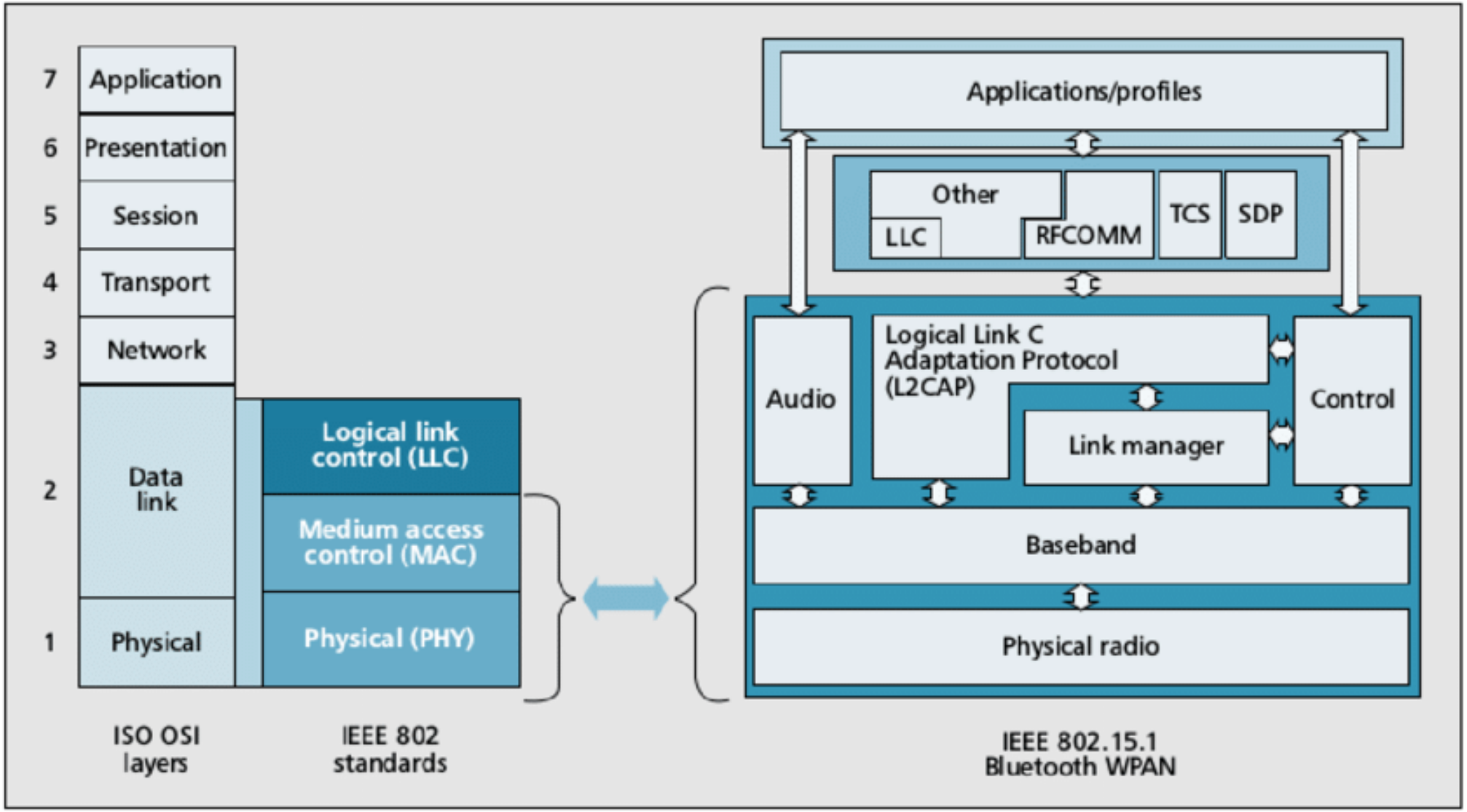


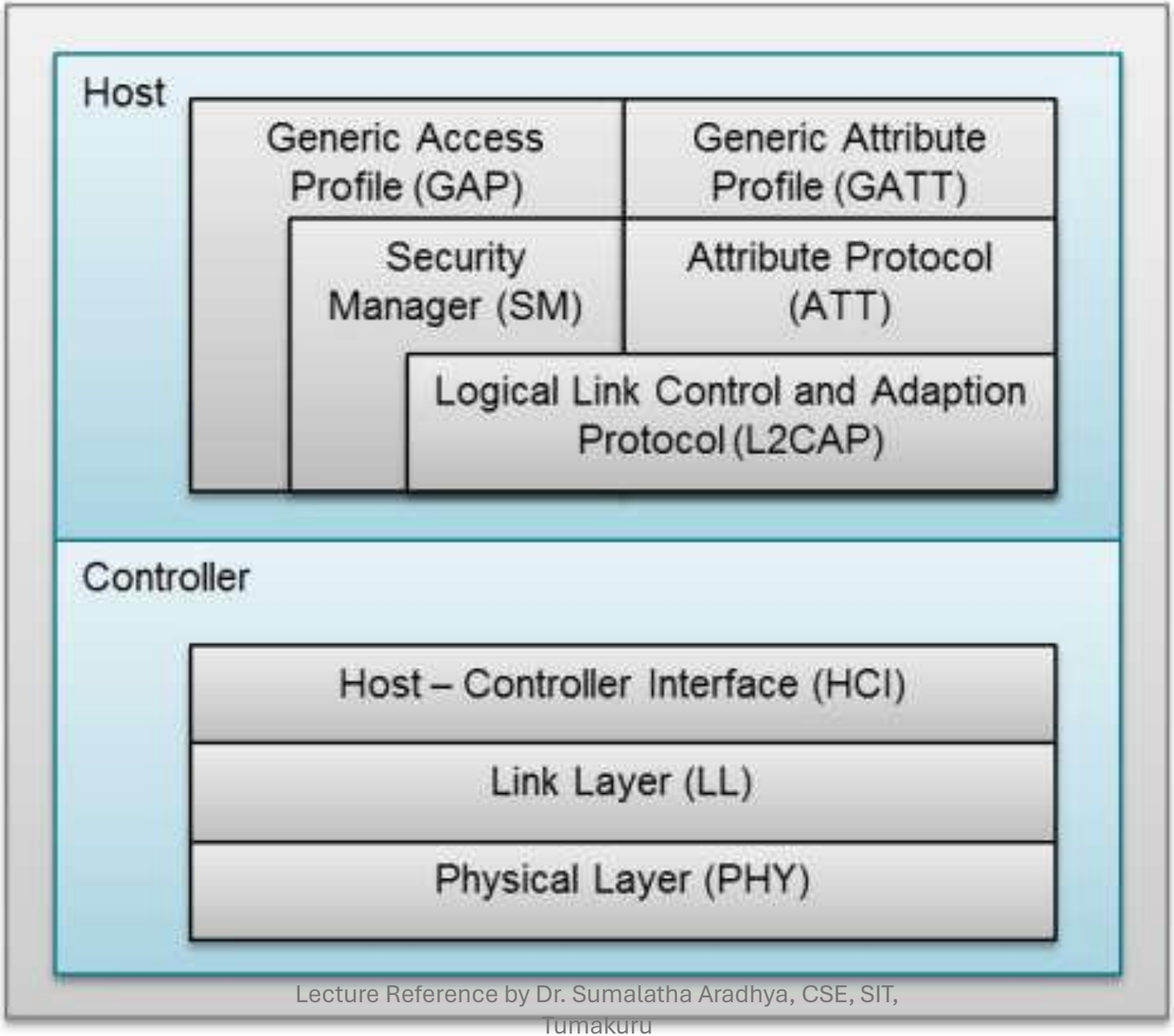
Image
Application
Client
Dispatch
ICall
Registration
Dispatch
Image
Server
Stack
Bluetooth low energy,
ZigBee or
Proprietary

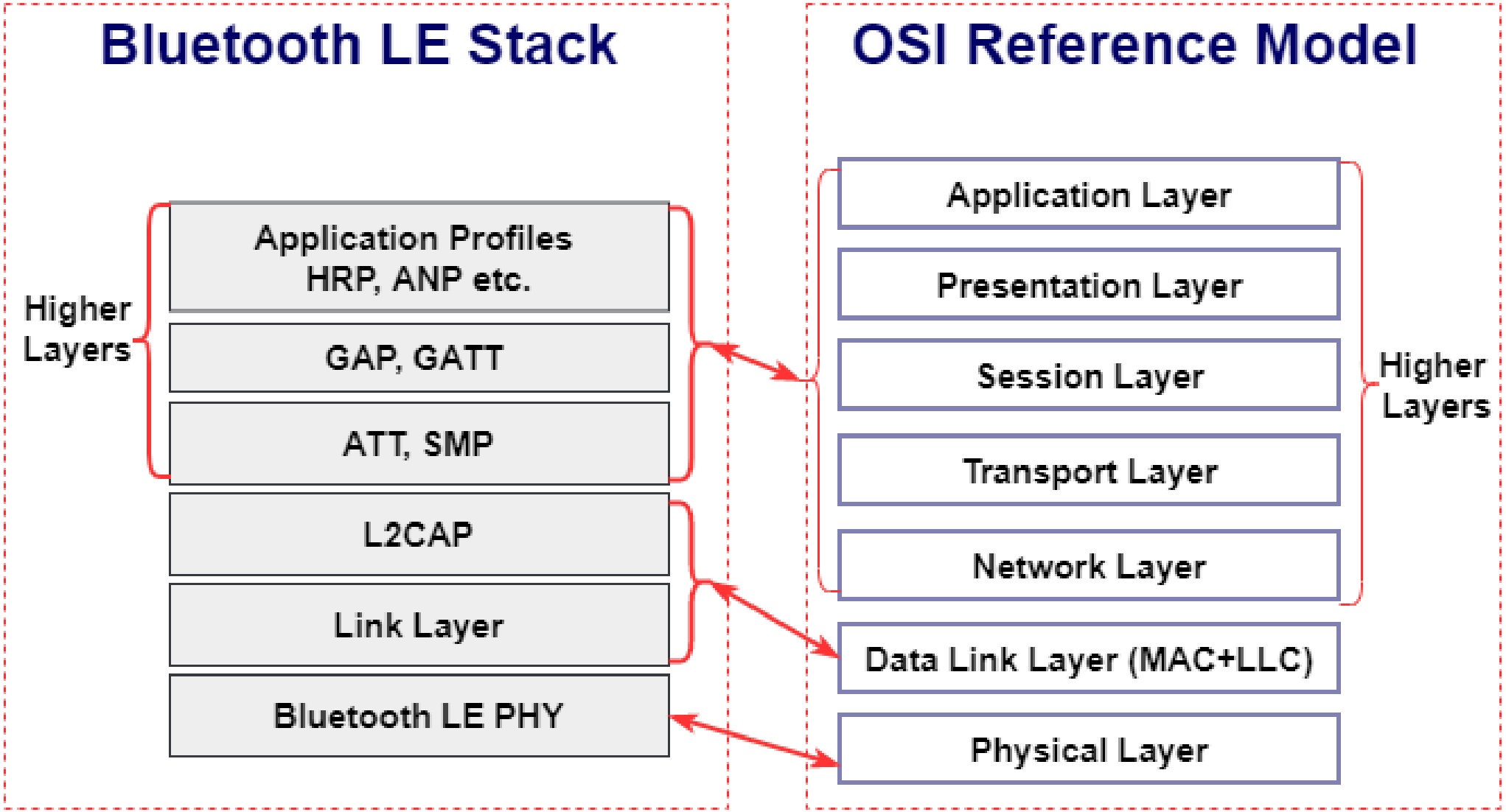
Bluetooth is a PAN technology based on IEEE 802.15.1. It is a specification for short-range wireless connectivity for portable personal devices initially developed by Ericsson. The Bluetooth SIG made their specifications publicly available in the late 1990s, at which time the IEEE 802.15 Group has took the Bluetooth work and developed a vendor-independent standard. The sublayers of IEEE 802.15:include: (i) RF layer; (ii) baseband layer; (iii) the link manager; and (iv) the L2CAP. Bluetooth has evolved through four versions; all versions of the Bluetooth standards maintain downward compatibility. BLE is a subset to Bluetooth v4.0 with an entirely new protocol stack for rapid build-up of simple links. BLE is an alternative to the "power management" features that were introduced in Bluetooth v1.0 to v3.0 as part of the standard Bluetooth protocols

(Bluetooth is a trademark of the Bluetooth Alliance, a commercial organization that certifies the interoperability of specific devices designed to the respective IEEE standard.)


Bluetooth Protocol Stack
Applications/Profiles
Audio
Other
L.L.C.
RFcomm
Telephony
Service discovery
Control
Logical link control adaptation protocol
Link manager
Baseband
Physical radio
Application layer
Middleware layer
Data link layer
Physical layer


7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data link
1 Physical
Logical link control (LLC)
Medium access control (MAC)
Physical (PHY)
ISO OSI layers
IEEE 802 standards
Applications/profiles
Other
LLC
RFCOMM
TCS
SDP
Audio
Logical Link C Adaptation Protocol (L2CAP)
Link manager
Control
Baseband
Physical radio
IEEE 802.15.1 Bluetooth WPAN


Host
Generic Access Profile (GAP)
Generic Attribute Profile (GATT)
Security Manager (SM)
Attribute Protocol (ATT)
Logical Link Control and Adaption Protocol (L2CAP)
Controller
Host – Controller Interface (HCI)
Link Layer (LL)
Physical Layer (PHY)


Bluetooth LE Stack
OSI Reference Model
Higher Layers
Application Profiles HRP, ANP etc.
GAP, GATT
ATT, SMP
L2CAP
Link Layer
Bluetooth LE PHY
Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer (MAC+LLC)
Physical Layer
Higher Layers

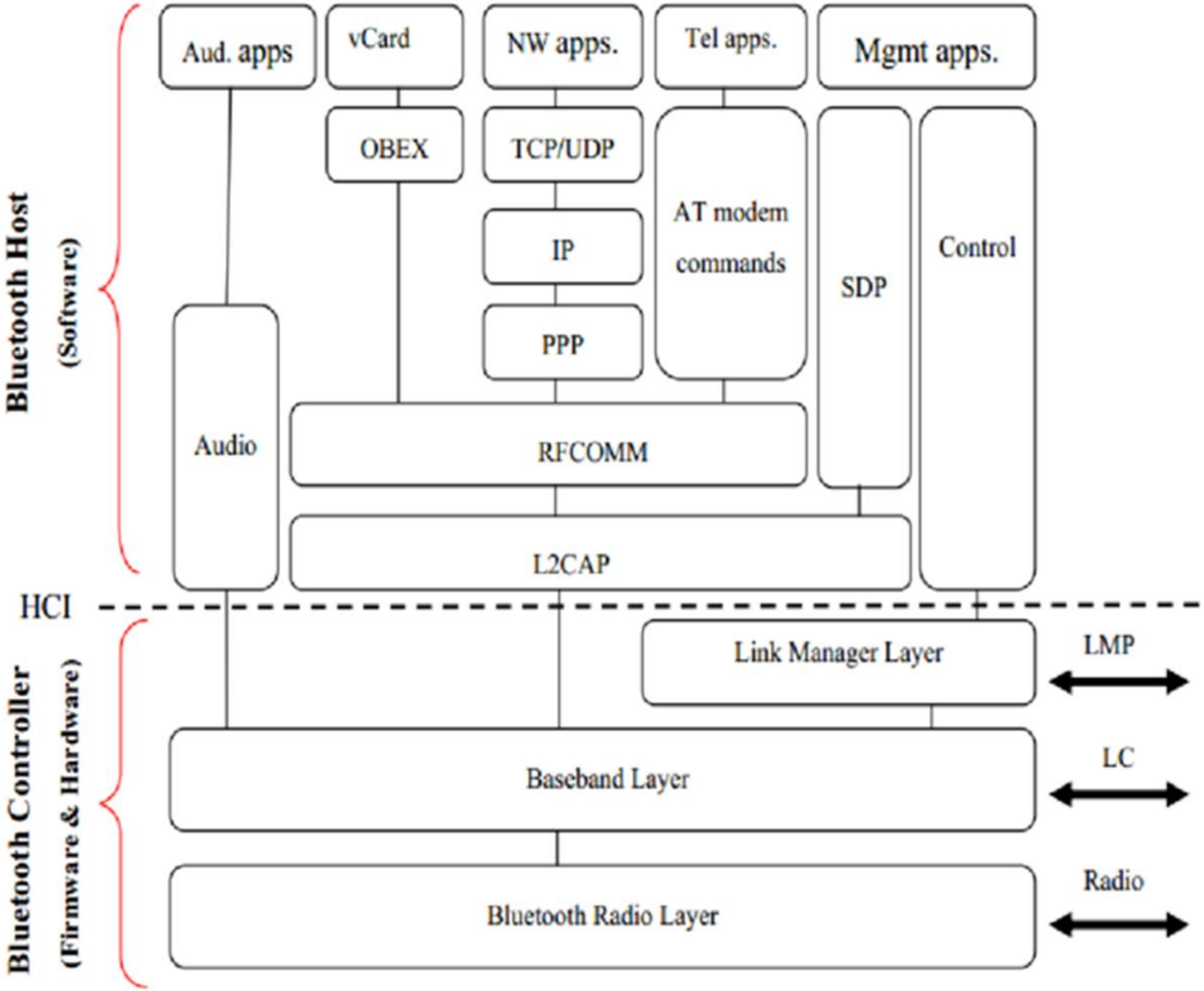


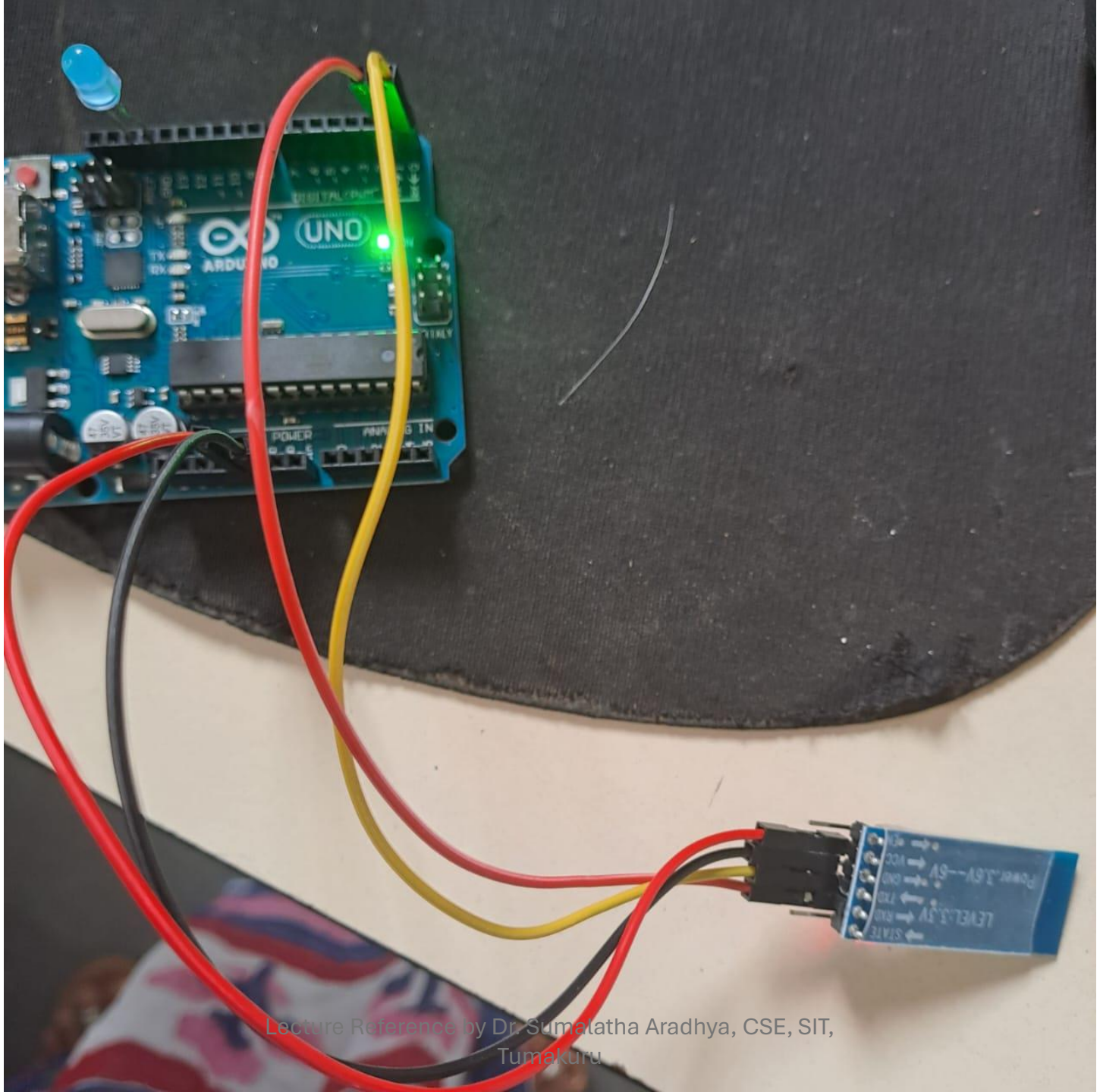
Figure Bluetooth protocol stack

Protocol stack implementation - bluetooth sdk

Lab experiment

Using Bluetooth module HC-05 , build application to

- Control LED on or off
- Send sensor data


UNO
ARDUINO
POWER

sketch_Bluetooth_lab5 | Arduino IDE 2.1.0

File Edit Sketch Tools Help

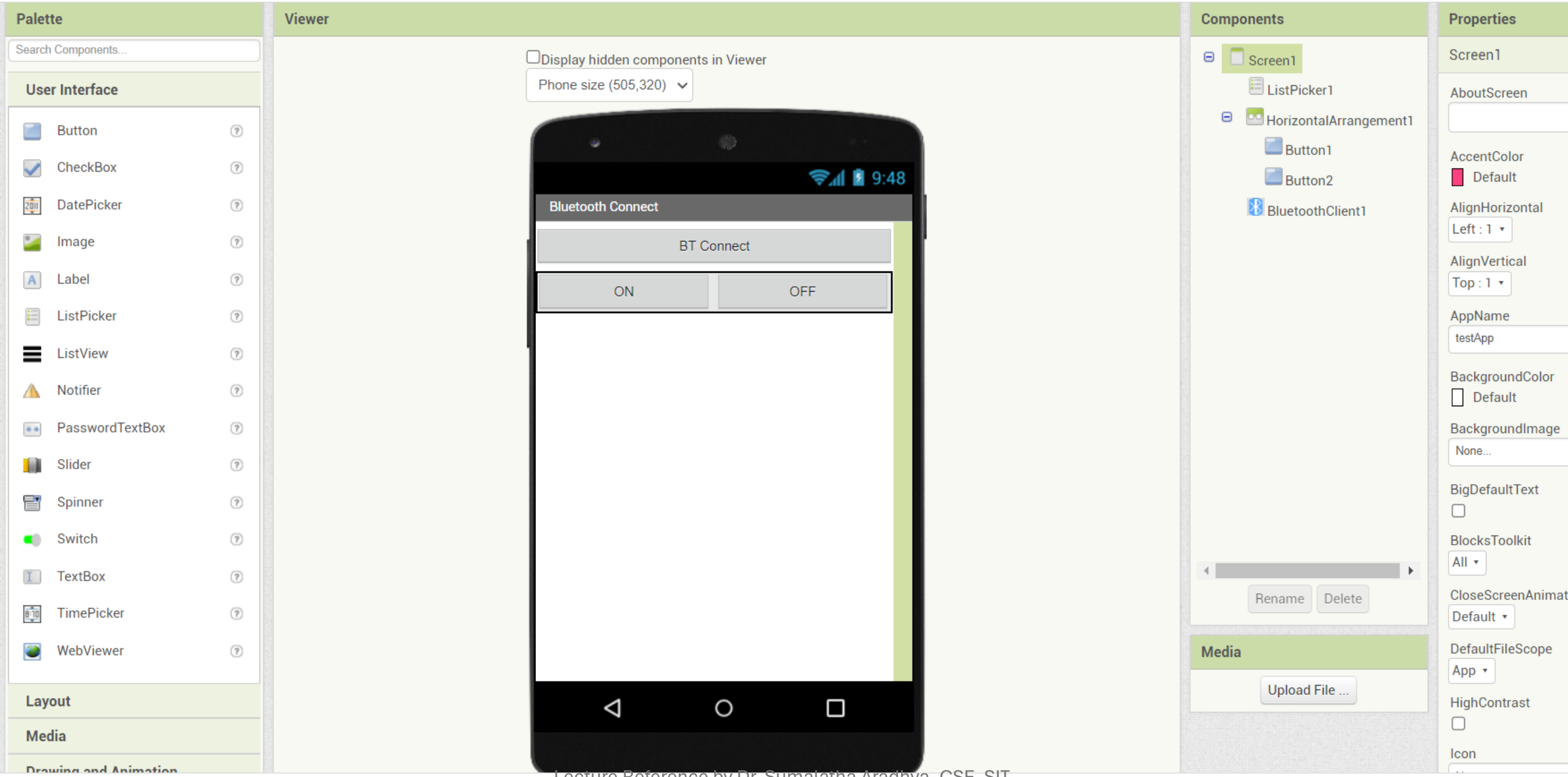
Arduino Uno

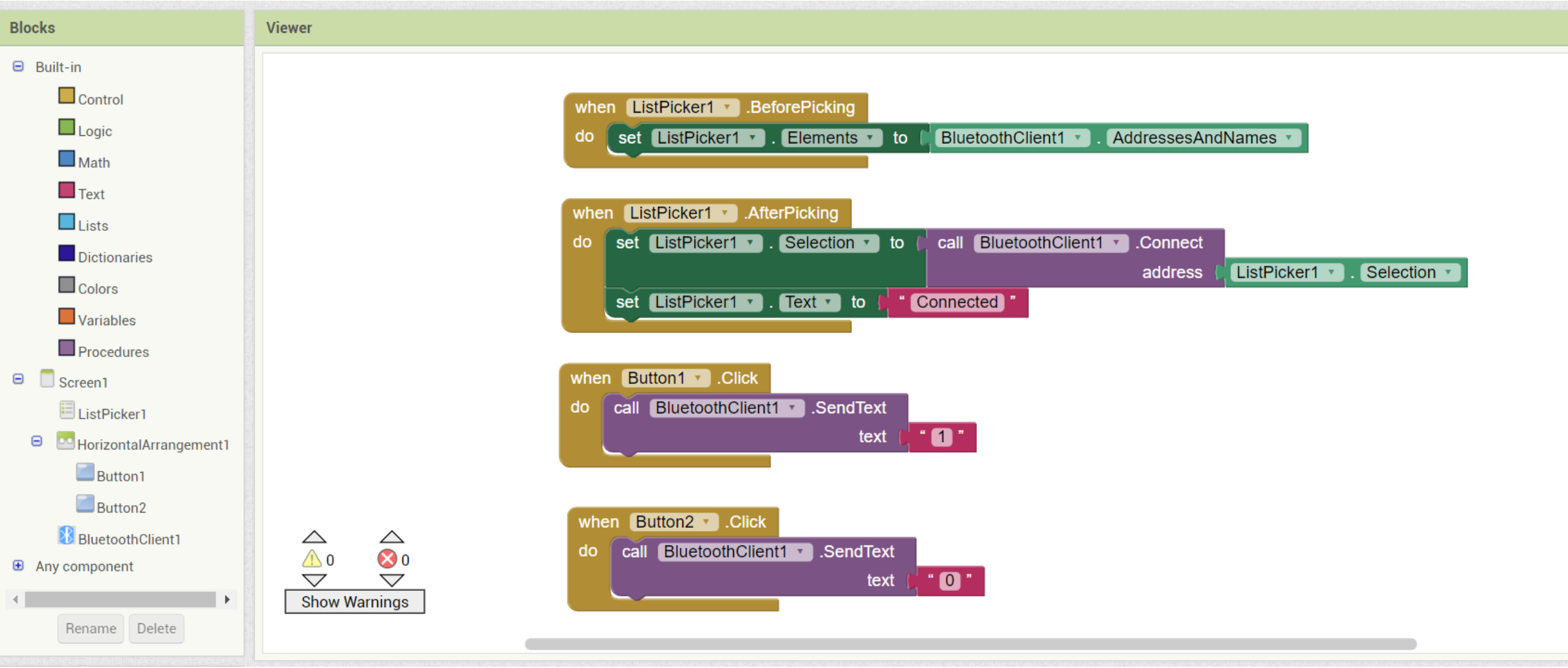
sketch_Bluetooth_lab5.ino

```
char bluetoothVal;   //value sent over bluetooth
char lastValue;      //stores the last value of the device
int ledpin=13;
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);     //begin communication on baud rate 9600
  pinMode(ledpin, OUTPUT); //set the led pin to OUTput pin mode
}
void loop() {
  // put your main code here, to run repeatedly:
  if(Serial.available())   //searches for available data
  {
    //if the value is received
    bluetoothVal = Serial.read();
  }
  if(bluetoothVal == '1')
  {
    digitalWrite(ledpin, HIGH);
    if(lastValue != '1')
    {
      Serial.println(F("LED ON"));
    }
    lastValue = bluetoothVal;
  }
  else if (bluetoothVal == '0')
  {
      digitalWrite(ledpin, LOW);
      if(lastValue != '0')
        Serial.println(F("LED OFF"));
    lastValue = bluetoothVal;
  }
  delay(1000);
}
```

Output Serial Monitor ×

While uploading do not connect BT module


Palette
Search Components...
User Interface
Button
CheckBox
DatePicker
Image
Label
ListPicker
ListView
Notifier
PasswordTextBox
Slider
Spinner
Switch
TextBox
TimePicker
WebViewer
Layout
Media
Viewer
Display hidden components in Viewer
Phone size (505,320)
9:48
Bluetooth Connect
BT Connect
ON
OFF
Components
Screen1
ListPicker1
HorizontalArrangement1
Button1
Button2
BluetoothClient1
Rename
Delete
Media
Upload File ...
Properties
Screen1
AboutScreen
AccentColor
Default
AlignHorizontal
Left : 1
AlignVertical
Top : 1
AppName
testApp
BackgroundColor
Default
BackgroundImage
None...
BigDefaultText
BlocksToolkit
All
CloseScreenAnimat
Default
DefaultFileScope
App
HighContrast
Icon



Build to generate .apk file, install .apk file @your mobile, Allow permission for App and BT connections, run app

sketch_Bluetooth_lab5 | Arduino IDE 2.1.0

File Edit Sketch Tools Help

Arduino Uno

sketch_Bluetooth_lab5.ino

```
    }

    if(bluetoothVal == '1')
    {
      digitalWrite(ledpin, HIGH);
      if(lastValue != '1')
      {
        Serial.println(F("LED ON"));
      }
      lastValue = bluetoothVal;
    }
    else if (bluetoothVal == '0')
    {
        digitalWrite(ledpin, LOW);
        if(lastValue != '0')
            Serial.println(F("LED OFF"));
        lastValue = bluetoothVal;
    }

    delay(1000);

```

Output Serial Monitor

Message (Enter to send message to 'Arduino Uno' on 'COM7')

```
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
```